

Project Stack and Sample Queries

Project Technologies Stack

1. Vector RAG Stack

- PDF Loader: PyMuPDF (fitz)
- Chunking: RecursiveCharacterTextSplitter (LangChain), 1000 chars / 200 overlap
- Embeddings: sentence-transformers/all-mpnet-base-v2 (local HuggingFace)
- Vector Store: FAISS (cached on disk)
- QA Model: Groq Llama-3.3-70b-versatile (via LangChain)

2. GraphRAG Stack

- Extraction: Frontier LLM for entity/relationship extraction
- Graph Store: Neo4j AuraDB (Cypher) and in-memory NetworkX
- Cypher Generation: LLM-generated Cypher
- Retrieval: Entity matching -> multi-hop traversal (depth ≤ 3) -> entity-to-chunk mention index -> evidence-density chunk ranking
- QA Model: Groq Llama-3.3-70b-versatile

3. General Dependencies & Tools

- LangChain Ecosystem: langchain, langchain-core, langchain-community, langchain-google-genai, langchain-openai, langchain-huggingface, langgraph
- Evaluation: ragas, trulens-eval
- Web App: streamlit
- Visualization: matplotlib, seaborn
- Data Processing: pandas, numpy, pydantic, python-dotenv, tiktoken

Sample Queries: GraphRAG vs VectorRAG Performance

1. Q4 (medium) - UltimateCreditor element

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A payment is credited to the account of a financing company, but the ultimate creditor of the payment is a customer of that financing company. Which element in the Credit Transfer Transaction Information block

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(Block C) is used to identify this customer as the ultimate beneficiary?

- Vector RAG (WRONG): "...the element used to identify the ultimate creditor (customer of the financing company) in the Credit Transfer Transaction Information block (Block C) is the 'Ultimate Creditor' element (2.149, 0..1). More specifically, within the 'Ultimate Creditor' element, the 'Name' (2.149, 0..1) and 'Identification' (2.151, 0..1) sub-elements are used to forward the name and ID of the ultimate creditor, respectively. Therefore, the correct answer is the 'Ultimate Creditor' element (2.149, 0..1)..." - wrong index (2.149 instead of 2.148).

- Graph RAG (CORRECT): "...the element used to identify the ultimate creditor (beneficiary) in the Credit Transfer Transaction Information block (Block C) is the 'Ultimate Creditor' element. According to the context, the 'Ultimate Creditor' element is defined as: 'The party who is the ultimate creditor of the payment. For example, a payment is credited to the account of a financing company, but the ultimate creditor is a customer of the financing company.' This element is part of the Credit Transfer Transaction Information block (Block C) and is used to identify the ultimate creditor of the payment, which in this case is the customer of the financing company." (it then appends the two source chunks, pages 7-8)

2. Q9 (hard) - Payment Status Report schema / status

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The payment message with MessageIdentification MSGID000001 passes structural validation at the bank. Which schema does the bank use for the return message, which Group Status does that return message carry, and which original identifier does it include?

Ground truth: The Payment Status Report schema pain.002.001.10 with Group Status ACTC; it returns the original MessageIdentification MSGID000001 (in the OrgnlMsgId element).

- Vector RAG (PARTIAL - hallucinated schema): "...the bank uses schema A: Structure, schema validation for the return message. According to the context on page 39, when the payment message is approved technically, the return message Group Status is ACTC, and the return message returns the value of the original MessageIdentification. Therefore, the bank uses the schema pain.002.001.10 return for the return

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message, the Group Status of the return message is ACTC, and the return message includes the original MessageIdentification, which is MSGID000001." - gets ACTC and MSGID000001 right but invents "schema A: Structure, schema validation".

- Graph RAG (CORRECT): "The bank uses the schema pain.002.001.10 for the return message (Source chunk | page 39). The return message carries the Group Status ACTC, indicating that the whole payment message is received and approved technically (Source chunk | page 39). The return message includes the original MessageIdentification, which is MSGID000001 (Source chunk | page 39)."